

Measuring multipartite entanglement through dynamic susceptibilities

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Entanglement is considered an essential resource in quantum technologies, and central to the understanding of quantum many-body physics. Developing protocols to detect and quantify the entanglement of many-particle quantum states is thus a key challenge for present experiments. Here, we show that the quantum Fisher information, a witness for genuinely multipartite entanglement, becomes measurable for thermal ensembles by means of the dynamic susceptibility—that is, with resources readily available in present cold atomic-gas and condensed-matter experiments. This establishes a connection between multipartite entanglement and many-body correlations contained in response functions, with immediate implications close to quantum phase transitions, where the quantum Fisher information becomes universal, allowing us to identify strongly entangled phase transitions with a divergent multipartite entanglement. We illustrate our framework using paradigmatic quantum Ising models, and point out potential signatures in optical-lattice experiments and strongly correlated materials.

Although entanglement is central to our understanding of quantum many-body physics¹, its direct measurement in an actual experiment to characterize quantum phases and phase transitions has remained elusive. In a few-body quantum system, in particular in a quantum information context involving few qubits, entanglement can be measured and quantified through tomography of the density matrix^{2–5}. In experiments with few atoms in optical lattices, such as engineered Bose– or Fermi–Hubbard models, entanglement entropy and purity of quantum states become measurable with protocols based on preparing copies of the quantum system in combination with the unique tools of atomic physics, including single-site manipulation and observation^{6,7}. However, an intrinsic feature of these protocols is an exponential scaling of resources, and thus an a priori limit to small system sizes.

In contrast, we will be interested below in quantifying multipartite entanglement for large system size, while still being accessible with experimentally realistic tools and resources. Our approach is based on measurement of the quantum Fisher information (QFI) as a witness of m -particle entanglement^{8–10}, as familiar from quantum metrology^{11–13}. As our main result, we will show that the QFI for thermodynamic equilibrium is directly related to measurement of dynamic susceptibilities (Fig. 1). To illustrate detection of entanglement by means of the QFI, we study several examples of quantum phase transitions (QPTs). We are able to identify a class of strongly entangled QPTs with divergent multipartite entanglement by studying the universal scaling of the QFI (refs 14–17), also at non-zero temperatures (see also refs 18–20 for scaling of related quantum metrics). The entanglement quantified through the QFI is not only a resource for quantum metrology^{8,9}, it also permits the characterization of exotic quantum effects such as many-body localization²¹. Although the outstanding feature of the presented protocol is its experimental simplicity, we will also discuss that the QFI as entanglement witness will be blind to non-local entanglement^{22,23}, such as, for example, in topological systems.

Background on the quantum Fisher information

Originally a concept from quantum metrology, the quantum Fisher information, F_Q , quantifies the maximal precision with which a parameter (a phase) ϑ can be estimated using a given quantum state ρ (refs 11–13). For M independent measurements, the variance of ϑ is bounded by $(\Delta\vartheta)^2 \geq 1/(MF_Q)$, the so-called quantum Cramér–Rao bound¹¹. A better precision—that is, a smaller right-hand side—can be reached if the state ρ is more sensitive towards a unitary transformation generated by the Hermitian operator $\hat{O}$ associated with ϑ . In other words, the QFI quantifies the distinguishability of ρ from $\rho' = e^{-i\vartheta\hat{O}}\rho e^{i\vartheta\hat{O}}$, for infinitesimal ϑ (without loss of generality, we set the mean value of ϑ to 0).

For a pure quantum state $\rho = |\psi\rangle\langle\psi|$, such as the ground state of a given Hamiltonian, the QFI assumes the simple form of a connected correlation function, which can be easily computed or measured,

$$F_Q = 4\Delta(\hat{O})^2 = 4(\langle\psi|\hat{O}\hat{O}|\psi\rangle - \langle\psi|\hat{O}|\psi\rangle^2) \quad (1)$$

Matters become much more complicated in a mixed state, such as a thermal ensemble $\rho = \sum_{\lambda} p_{\lambda} |\lambda\rangle\langle\lambda|$, where $|\lambda\rangle$ is the energy eigenbasis with occupation probabilities $p_{\lambda} = \exp(-E_{\lambda}/T)/Z$, with Z the partition function. In such a case, the QFI takes the considerably more complex structure

$$F_Q = 2 \sum_{\lambda, \lambda'} \frac{(p_{\lambda} - p_{\lambda'})^2}{p_{\lambda} + p_{\lambda'}} |\langle\lambda|\hat{O}|\lambda'\rangle|^2 \quad (2)$$

(where the sum includes only terms with $p_{\lambda} + p_{\lambda'} > 0$).

Importantly, the quantum Cramér–Rao bound can fall below classical limits if ρ describes an entangled state of $N > 1$ particles¹³. Consider a local generator $\hat{O} = \sum_{i=1}^N \hat{O}_i$, where $\hat{O}_i$ has a spectrum of unit width (see Methods). If ρ achieves a sufficiently large QFI to break classical bounds,

$$f_Q \equiv F_Q/N > m \quad (3)$$

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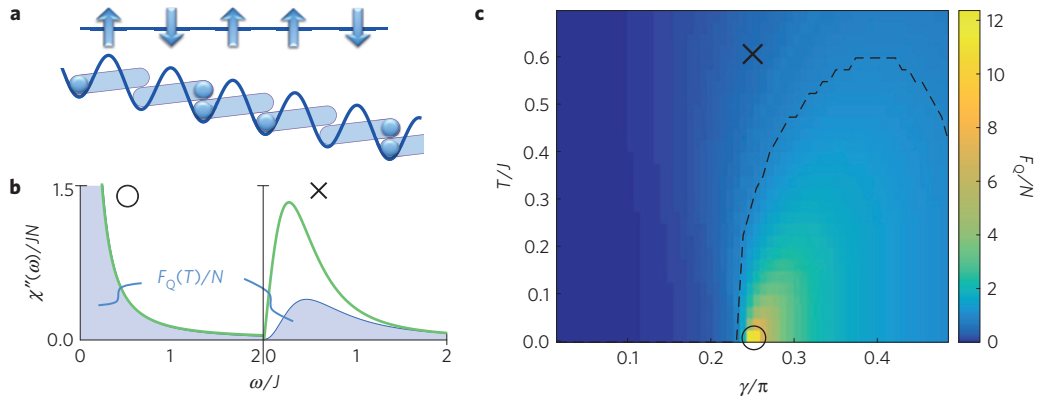


Figure 1 | Measurement prescription for the quantum Fisher information (QFI), a witness for multipartite entanglement. **a**, A quantum many-body system is prepared in a thermal state at temperature T . The example shows an Ising spin chain (top), equation (8), describing solid-state samples^{32,33} and optical-lattice experiments (bottom)³¹. The concept applies also to fermions and bosons, and in any spatial dimension. **b**, Standard tools, such as inelastic Bragg or neutron scattering, measure the imaginary part of the dynamic susceptibility $\chi''(\omega, T)$ as a function of frequency exchange ω [green line, computed using equation (11)]. The integral $F_Q(T) = (4/\pi) \int_0^\infty d\omega \tanh((\omega/2T)) \chi''(\omega, T)$ gives the QFI (shaded areas). **c**, This procedure allows the QFI to be mapped out as a function of temperature and transverse field, parametrized by γ (exact data for $N=64$). Low-temperature states (circle) can host entanglement, but it is lost at higher temperatures (cross). The quantum Ising chain has divergent entanglement close to the quantum critical point, and a robust entangled region extending to finite temperatures (enclosed by the dashed line).

with f_Q the QFI density and m a divisor of N , then ρ must be $m+1$ -partite entangled^{8,9}.

Recently, in a remarkable atomic-gas experiment¹⁰, it has been demonstrated that a lower bound on the QFI can be measured by studying the behaviour of an observable's probability distribution under the unitary transformation $e^{i\hat{O}}$. This could then be used to demonstrate the presence of bipartite entanglement. Here, we show how the QFI can be measured directly and efficiently for any quantum many-body system in a thermal state at any temperature.

Main result

As the major result of this work, we rigorously relate the QFI to a Kubo response function,

$$F_Q(T) = \frac{4}{\pi} \int_0^\infty d\omega \tanh\left(\frac{\omega}{2T}\right) \chi''(\omega, T) \quad (4)$$

where $\chi''(\omega, T) = \Im(\chi(\omega, T))$ is the imaginary, dissipative part of the dynamic susceptibility in the state ρ with respect to $\hat{O}$ —the same thermal state and generator for which the QFI is evaluated. A similar connection has already been noted for the related geometric tensor in the ground state^{18,19}, and response functions in imaginary time are being used to make the geometric tensor calculable in quantum Monte Carlo computations (see, for example, ref. 24). Equation (4) relates the QFI to response functions at non-zero temperatures and in real time, thus providing a tool to measure it directly in laboratory experiments.

Proof: The proof of equation (4) is straightforward. It requires only the minimal assumption of thermal equilibrium. Setting $\hbar = 1 = k_B$, the dynamic susceptibility is defined as

$$\chi(\omega, T) = i \int_0^\infty dt e^{i\omega t} \text{tr} \left(\rho [\hat{O}(t), \hat{O}] \right) \quad (5)$$

where $\hat{O}(t) = e^{iHt} \hat{O} e^{-iHt}$. It is convenient to work in the Lehmann representation, that is, the energy eigenbasis, where

$$\chi''(\omega) = \sum_{\lambda, \lambda'} (p_\lambda - p_{\lambda'}) |\langle \lambda | \hat{O} | \lambda' \rangle|^2 \pi \delta(\omega - E_{\lambda'} + E_\lambda) \quad (6)$$

Exploiting that for a thermal state $\int_{-\infty}^\infty d\omega \tanh(\omega/2T) \delta(\omega - E_{\lambda'} + E_\lambda) = \tanh((E_{\lambda'} - E_\lambda)/2T) = (p_\lambda - p_{\lambda'})/(p_\lambda + p_{\lambda'})$, using the

asymmetry of $\chi''(\omega, T)$, and correcting for prefactors, we directly obtain equation (4). This proof can be straightforwardly extended to the QFI matrix and, through the fluctuation–dissipation theorem, to the dynamic structure factor $S(\omega)$. Equation (4) also presents some direct corollaries, such as a sum rule which we discuss in the Supplementary Methods.

The identification given by equation (4) has several conceptual implications.

First, it makes the QFI, a witness for multipartite entanglement, a straightforwardly measurable quantity. Dynamic susceptibilities are routinely measured in many-body systems using well-established techniques such as Bragg spectroscopy^{25,26} or neutron scattering²⁷.

Second, the central equation (4) has also theoretical implications. As one example, the dynamical susceptibility measures quantum fluctuations. Because fluctuations determine the sensitivity of a state towards external perturbations, the existence of a relation to the QFI is intuitive, which now becomes rigorous using equation (4). From a different point of view, the QFI extracts the entanglement content of the quantum correlations contained in $\chi''(\omega, T)$.

Third, the connection provided by equation (4) has direct quantitative consequences near continuous QPTs when choosing for $\hat{O}$ a relevant operator in the renormalization-group sense, such as the order parameter. Then, known universal scaling laws for $\chi''(\omega, T)$ translate directly into universal scaling for F_Q . In magnetic materials, for example, experiments have observed the scaling $\chi''(\omega, T) \sim (T/J)^{-s} \phi(\omega/T)$, with ϕ a universal function and s a scaling exponent (see, for example, refs 28,29). Integrating this, one directly obtains the scaling $f_Q \sim (T/J)^{1-s}$. Universal scaling has already been observed theoretically for the QFI in ground states of many-body models^{14–17}, and has been characterized for the geometric tensor in the ground state^{18,19}, as well as thermal states²⁰. In the following, we discuss the general scaling behaviour of the QFI in the experimentally relevant regime of non-zero temperatures.

Universal scaling of multipartite entanglement

Consider a local generator $\hat{O} = \sum_{i=1}^N \hat{O}_i$, in a d -dimensional system with linear size L , lattice spacing $a = 1$, and $N = L^d$ sites. As explained in the Supplementary Methods, the universal behaviour of f_Q , following standard scaling arguments, is

$$f_Q(T/J, L^{-1}, \tilde{h}) = \lambda^{\Delta_Q} \phi_Q(\lambda^z T/J, \lambda L^{-1}, \lambda^{1/\nu} \tilde{h}) \quad (7)$$

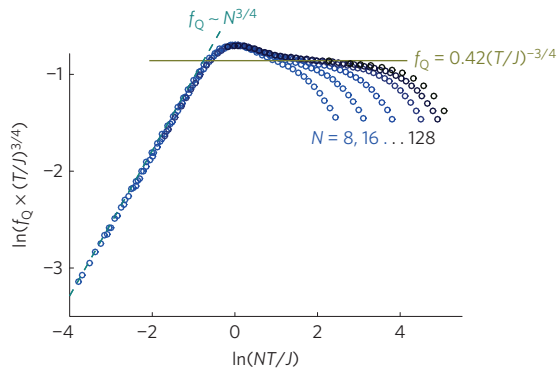


Figure 2 | Universal scaling of the quantum Fisher information density, $f_Q = F_Q/N$, calculated for the order parameter in the quantum Ising chain. Exact data at the critical point and for $N = 8, 16, 32, 64, 96, 128$. At low temperatures, the universal scaling laws produce an excellent data collapse. In a range of intermediate temperatures, the scaling approaches analytic predictions for the thermal critical regime in the thermodynamic limit³⁴ (solid line). The strong divergence at small temperatures of $f_Q \sim N^{3/4}$ (dashed line) implies a diverging multipartiteness of entanglement.

Here, z is the dynamical and ν the correlation-length critical exponent. λ is the cutoff scale determined by the relevant perturbations L^{-1} , T , and the normalized distance from the critical point $\tilde{h}$. The scaling dimension of f_Q is $\Delta_Q = d - 2\Delta_\alpha$, with Δ_α the scaling dimension of $\hat{O}_i$.

Because f_Q bounds the number of entangled particles according to equation (3), its scaling behaviour allows us to identify a class of strongly entangled QPTs—that is, QPTs with a divergent multipartiteness of entanglement. These are those transitions with $\Delta_Q > 0$. The scaling behaviour also implies a length scale l_{ent} (ref. 30) over which multipartite entanglement as measured by the QFI exists, $l_{\text{ent}} \sim f_Q^{1/d} \sim \lambda^{1-2\Delta_\alpha/d}$ (see Supplementary Methods).

The framework described up to now is completely general. To illustrate its viability for understanding entanglement in quantum many-body systems, we now turn to specific examples. We focus on a paradigmatic class of model systems exhibiting a QPT, namely Ising models in a transverse field, which are realizable in quantum-optical^{10,31} as well as solid-state systems^{32–34},

$$\frac{H}{J} = -\cos(\gamma) \sum_{l,j=1}^N \mathcal{J}_{lj} \sigma_l^x \sigma_j^x + \sin(\gamma) \sum_{l=1}^N \sigma_l^z \quad (8)$$

Here, σ_l^α , $\alpha = x, y, z$, is the Pauli matrix on lattice site l . Depending on the interactions $\mathcal{J}_{lj}$, this Hamiltonian exhibits a quantum critical point at some critical field strength γ_c . The order parameter for the transition is $\sum_l \langle \sigma_l^x \rangle / N$. Its strong critical fluctuations make the generator $\hat{O} = \sum_l \sigma_l^x / 2$ an ideal candidate for testing the scaling behaviour of the QFI.

We first focus on the simplest case, the one-dimensional nearest-neighbour Ising chain, $\mathcal{J}_{lj} = \delta_{l,j+1}$, where the scaling exponents are known analytically, $z = 1$ and $\Delta_\alpha = 1/8$ (ref. 34). This gives $\Delta_Q = 3/4$; that is, the nearest-neighbour quantum Ising chain lies in the class of strongly entangled phase transitions with divergent multipartite entanglement. Indeed, f_Q for the order parameter exhibits a strong peak around the critical point $\gamma_c = \pi/4$ (see Fig. 1c). The entanglement radiates out from the peak, generating a broad entangled region also at non-zero T (refs 35,36).

To illustrate the Ansatz described by equation (7), we consider the scaling with system size L and temperature T , at fixed $\tilde{h} = 0$. For $L \ll (T/J)^{-1/z}$, the dominant cutoff scale is $\lambda \sim L = N$, implying

$$f_Q \sim N^{3/4} \quad (9)$$

The data in Fig. 2 reproduces perfectly this strong algebraic growth, which is remarkably close to the theoretical maximum of $f_Q = N$. The associated multipartite entanglement length scale is thus highly divergent, $l_{\text{ent}} \sim N^{3/4}$.

With increasing temperature, the cutoff scale crosses over to $\lambda \sim (T/J)^{-1/z}$, and the scaling becomes $f_Q = C(T/J)^{-3/4}$. The constant $C \approx 0.42$ can be obtained using analytical results for the dynamic susceptibility at criticality in the thermodynamic limit³⁴ (see Methods). In the temperature regime of validity, $NT/J \gg 1$ and $T/J \ll 1$, the exact data for finite chains is consistent with this scaling prediction (see Fig. 2). For $T \gg J$, the system crosses over into a generic high-temperature asymptotic behaviour $f_Q \sim (T/J)^{-2}$.

Absence of signature at thermal phase transitions

Remarkably, scaling behaviour is observed only at quantum, but not thermal phase transitions, because equation (4) considers only quantum fluctuations. A simple example to demonstrate the insensitivity towards thermal phase transitions is provided by the fully connected transverse-field Ising model, $\mathcal{J}_{lj} = 1/N$, $\forall l, j$ (refs 15,37), similar to the model describing the experiments of ref. 10. In contrast to its nearest-neighbour counterpart, this model also exhibits, in addition to the QPT at $\gamma_c = \pi/4$, a thermal phase transition³⁸ (see Methods).

Figure 3a shows f_Q for the order parameter in the temperature–transverse-field plane. We delegate its scaling analysis to the Supplementary Methods. More important at this point, whereas f_Q shows a divergence at the QPT, no particular feature can be discerned at the thermal phase transition. Neither do such features appear in derivatives of f_Q (Fig. 3b).

However, it is known that the static isothermal susceptibility χ^T of the order parameter diverges³⁹. The reason why the QFI instead remains featureless becomes clear when decomposing $\chi^T = \chi_{\text{el}} + \chi_{\text{v,v}}$ into its two fundamental parts⁴⁰, the elastic (or Curie) contribution χ_{el} , and the quantum-mechanical von Vleck correction $\chi_{\text{v,v}}$, which is continuously connected to the Kubo susceptibility, $\chi_{\text{v,v}} = \lim_{\omega \rightarrow 0} \chi(\omega, T)$. It is χ_{el} that diverges at a thermal phase transition. As is shown in the Supplementary Methods, χ_{el} can be related to the Fisher information in a classical scenario⁴¹ that has no relation to entanglement. The second term $\chi_{\text{v,v}}$, on the other hand, remains smooth at thermal transitions. The QFI, thus, considers only the contribution to the susceptibility that is due to quantum fluctuations, and remains insensitive to thermal phase transitions.

Experimental considerations

Let us finally address some practical aspects that will be important for experiments. The measurement prescription given by equation (4) is very flexible, as its proof did not make any assumptions on microscopic details of the system under study (other than thermal equilibrium). As a consequence, it applies in any spatial dimension, for any Hermitian generator $\hat{O}$, and it can be equally used for systems of spins, bosons, or fermions. A bound for multipartite spatial entanglement as in equation (3) is known to exist in all cases where the generator is a sum of local operators, $\hat{O} = \sum_l \hat{O}_l$, when $\hat{O}_l$ has a bounded spectrum¹³ (see Methods). In this context, it is important to note that the scaling analysis overcomes a usual practical difficulty for studying the QFI, the optimal choice for the linear generator. In the vicinity of a QPT, the choice becomes clear: one may select any suitable, relevant operator $\hat{O}_l$; preferably the one with the largest scaling exponent Δ_Q . A good choice will often be the order parameter, as in the examples above. The Supplementary Methods contains an example for the Mott-insulator–superfluid transition, where universal behaviour is extracted from an operator different from the order parameter that is accessible by Bragg spectroscopy. Importantly, once an appropriate generator is chosen, the complexity of measuring

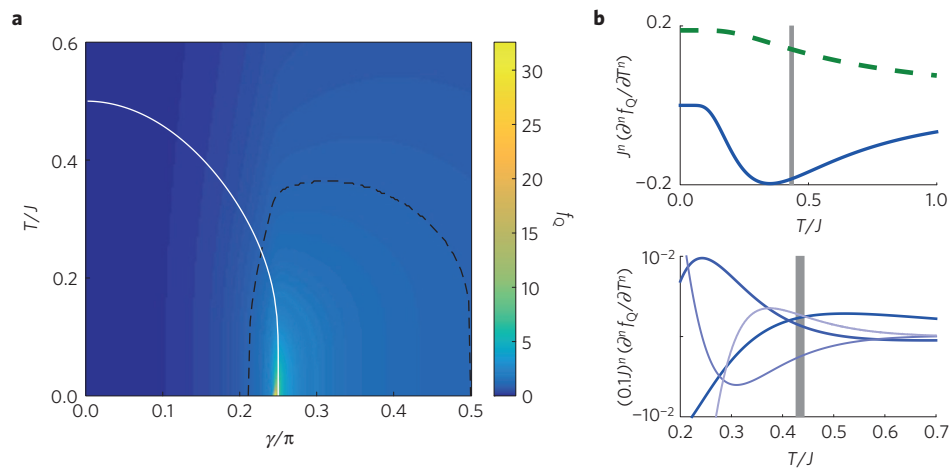


Figure 3 | Absence of features of the quantum Fisher information at thermal phase transitions, exemplified by the infinitely connected Ising model.

a, Finite entanglement is witnessed at non-zero temperatures (enclosed by the dashed line), and a divergence appears near the quantum critical point at $\gamma = \pi/4$ and $T = 0$. But, crucially, f_Q shows no features at the thermal phase transition (solid line). Exact data for $N = 1,000$. **b**, Derivatives of f_Q also remain smooth at the thermal transition (vertical line). Top: f_Q (dashed) and $\partial f_Q / \partial (T/J)$ (solid line). Bottom, from dark to light: second to fifth derivatives. Exact data for $N = 1,600$ and $\gamma = \pi/8$.

$\chi''(\omega, T)$ does not scale with system size, thus allowing for an efficient evaluation of the QFI by means of equation (4).

Remarkably, the measurement prescription is robust against intrinsic sources of imperfections. For example, an uncertainty in determining the system's temperature can be mitigated by choosing the largest temperature estimate. Owing to the monotonicity of the $\tanh(\omega/2T)$ as a function of frequency, this produces a lower bound for the QFI. Another natural error source is a finite spectral resolution. Assuming a symmetric broadening function (such as the Gaussians commonly used in neutron spectroscopy²⁷), the concavity of the $\tanh$ assures that the integration again returns a lower bound to the true QFI, with less severe consequences at smaller temperatures. Also, the integral in equation (4) will be limited to some frequency range (the upper integral limit represents, as usual, a scale much larger than energies accessible by the considered degrees of freedom). This limitation will again deliver a lower estimate. Remarkably, none of these errors will produce a false positive indicator for entanglement. The integration in equation (4) thus entails a certain intrinsic robustness, as compared to more error-prone measurements at precisely one fixed ω . What is more, the existence of the universal scaling laws assures an additional inherent robustness of the QFI close to a QPT. Therefore, in contrast to some engineered highly entangled states, such as the Greenberger–Horne–Zeilinger (GHZ) state, the entanglement witnessed by the QFI close to QPTs is unaffected by weak symmetry-preserving perturbations.

The dynamic susceptibility has been studied experimentally in a wide variety of strongly correlated materials (see, for example, refs 28,29,32,42–44). It will be worthwhile to revisit such experiments in light of equation (4). Although it is challenging in neutron spectroscopy to measure absolute values, they can be obtained by appropriate normalization, for example, through incoherent elastic or phonon scattering⁴⁵, allowing neutron scattering with current instrumentation to account for 99(8)% of the spectral weight⁴⁶. Moreover, the general scaling behaviour of multipartite entanglement may already be estimated without such precise normalization, and even from existing data. For example, measurements in certain quantum-critical magnetic materials are consistent with a scaling exponent $\Delta_Q > 0$ (refs 28,29), as we require for a divergent QFI. Scaling behaviour for dynamic susceptibilities can sometimes even be observed in the absence of QPTs, such as in certain heavy-fermion compounds at elevated temperatures⁴².

Prospects

The proposed measurement procedure is scalable to large particle numbers, and is thus complementary to exponentially expensive approaches such as refs 6,7. Whereas the latter can in principle access non-local entanglement, as characterizes, for example, topological phases^{22,23}, physical susceptibilities are related to local operators. Hence, although the QFI does exhibit scaling behaviour at topological transitions, it does not identify topological phases, as we illustrate in the Supplementary Methods using the example of the Kitaev wire. It shares this fate with other entanglement witnesses^{1,47} (for some recent applications to quantum many-body experiments, see refs 48–51). A remaining challenge is the question whether non-local extensions of the proposed protocol permit the characterization of topological entanglement²³ and exotic quantum phases^{22,43}.

The multipartite entanglement that the QFI detects has an immediate interpretation as a resource for quantum metrology^{8,9}. Further potential applications range from quantifying the entanglement in quantum simulators of many-body problems^{52–54} to the characterization of strongly correlated systems. For example, a natural question is whether entanglement influences such striking material properties as high-temperature superconductivity, where scaling behaviour has already been observed in the optical conductivity⁵⁵. Specifically, an underlying antiferromagnetic critical point⁴⁴ will show quantum-critical scaling of entanglement, which may also be tested in cold atomic-gas experiments^{56,57}.

Methods

Methods and any associated references are available in the [online version of the paper](#).

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Author contributions

All authors devised the project, discussed the results, and wrote the manuscript.

Additional information

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Competing financial interests

The authors declare no competing financial interests.

Methods

Relationship of the QFI to multipartite entanglement. The QFI has a deep connection with multipartite entanglement^{8,9}. Consider a system of N spins with length $S=1/2$, and a linear observable $\hat{O}=(1/2)\sum_i \mathbf{n}_i \cdot \boldsymbol{\sigma}_i$, where $\boldsymbol{\sigma}_i=(\sigma_i^x, \sigma_i^y, \sigma_i^z)$ is a vector containing the Pauli matrices σ_i^α associated with spin i , and $\mathbf{n}_i=(n_i^x, n_i^y, n_i^z)$ is a unit vector on the Bloch sphere. Then the system hosts at least $m+1$ -partite entanglement if the QFI associated with $\hat{O}$ fulfils⁸

$$F_Q > \left\lfloor \frac{N}{m} \right\rfloor m^2 + \left(N - \left\lfloor \frac{N}{m} \right\rfloor m \right)^2 \quad (10)$$

where $\lfloor X \rfloor$ is the largest integer smaller than or equal to X . For m a divisor of N , the condition in equation (10) attains the simple form of equation (3) when expressed through the QFI density $f_Q \equiv F_Q/N$. Note that, typically, response functions that measure linear operators (at fixed wavevector transfer q) contain a factor of $1/N$ relative to our definition, equation (5), so they will observe f_Q directly rather than F_Q .

The proof of ref. 8 for bounding the multipartite entanglement with equation (10) can be directly translated to degrees of freedom other than spin $1/2$, as long as $\hat{O}$ represents a sum of local operators with a bounded spectrum¹³. If $h_{\max}$ and $h_{\min}$ denote the largest, respectively smallest, eigenvalue of $\hat{O}$, then the right-hand side of equation (10) acquires the prefactor $(h_{\max} - h_{\min})^2$. Therefore, the QFI can also witness spatial entanglement in systems other than spin $1/2$, such as larger spins. Note that the relation of the QFI to response functions, equation (4), is independent of any such microscopic details of the underlying quantum many-body system or the Hermitian operator $\hat{O}$, which may even be non-local or unbounded.

Solvability of the considered models. All models used in this article for illustrating the main concepts are exactly solvable (throughout we set $\hbar=k_B=a=1$, with a the lattice spacing). The one-dimensional Ising chain in a transverse field [equation (8) with $\mathcal{J}_{ij}=\delta_{j,i+1}$] can be mapped to a free-fermion problem³⁴. Dynamical susceptibilities for finite systems can then be calculated by means of Wick's decomposition of expectation values⁵⁹, and the Pfaffians appearing in the resulting expressions can be evaluated efficiently using the algorithm

described in ref. 58. Exact results for finite systems with the order parameter as generator are shown in Figs 1c and 2.

Moreover, for the transverse Ising chain in the thermodynamic limit, analytical results exist for the order-parameter dynamic susceptibility in the quantum-critical region³⁴. For wavevector transfer $k=0$, at non-zero temperatures above the critical point ($\tilde{h}=0$) the dynamic susceptibility reads

$$\lim_{N \rightarrow \infty} \frac{\chi(\omega)}{N} = c \frac{J^{3/4}}{T^{7/4}} \left(\frac{\Gamma\left(\frac{1}{16} - i\frac{\omega}{4\pi T}\right)}{\Gamma\left(\frac{15}{16} - i\frac{\omega}{4\pi T}\right)} \right)^2 \quad (11)$$

where $c \approx 3.81 \times 10^{-3}$ and Γ denotes the Gamma function. Performing the ω -integral in equation (4), one immediately obtains the QFI,

$$f_Q(T) = C (J/T)^{3/4} \quad (12)$$

with $C \approx 0.42$. Data based on these equations is presented in Fig. 1b, and as one limiting case in Fig. 2.

The infinite-range Ising Hamiltonian [equation (8) with $\mathcal{J}_{ij}=1/N$] commutes with both $\mathbf{S}^2=(S^x)^2+(S^y)^2+(S^z)^2$ and S^z , where $S^\alpha=\sum_i \sigma_i^\alpha/2$, $\alpha=x,y,z$. As a consequence, the Hamiltonian decomposes into disconnected blocks when represented in the common eigenbasis of $\mathbf{S}^2$ and S^z . Each block grows linearly with particle number N , and can be diagonalized efficiently, allowing one to compute the dynamical susceptibility exactly, even for large systems. Here, we consider the largest of these blocks, with dimension $N+1$. Figure 3 presents corresponding exact data. The infinite-range Ising model has a thermal phase transition for $\gamma < \pi/4$, with critical temperature³⁸

$$\frac{T_c}{J} = \frac{\sin(\gamma)}{\log[(1+\tan(\gamma))/(1-\tan(\gamma))]} \quad (13)$$

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In this supplementary material, (1) we discuss the relationship Fisher-information/susceptibility in a classical, thermal setting; (2) we present a sum rule to bound the quantum Fisher information; (3) we explain in detail the scaling behaviour for the quantum Fisher information; (4) we present additional numerical results for universal scaling: (4.A) a quantum phase transition above the upper critical dimension in the infinite-range Ising model, (4.B) a transition coinciding with a factorization point in one-dimensional hard-core bosons in an optical lattice, and (4.C) a symmetry-protected topological transition in the Kitaev wire.

S1. CLASSICAL CONTRIBUTION TO QFI AND THERMAL PHASE TRANSITIONS

In the main text, we study how well thermal states that lie at close points in a phase diagram are distinguishable, focusing entirely on quantum effects. More precisely, we ask how fast the probability distribution of a thermal state $\rho = \sum_{\lambda} p_{\lambda} |\lambda\rangle \langle \lambda|$, with $p_{\lambda} = \exp(-E_{\lambda}/T)/Z$, changes due to the unitary transformation with $e^{i\vartheta\hat{O}}$. It is instructive to extend this analysis to any source of distinguishability, be it quantum or classical.

One may extend the present scenario by coupling the system to an external bath and including a sufficiently long waiting time, such that the populations can rethermalise. The changing parameter ϑ may then signify the temperature. The change of populations that this induces provides a second source of distinguishability between ρ and ρ' , which to leading order remains disparate from the one described above. We can incorporate this effect by considering the full quantum Fisher information (QFI) [1, 2]

$$\begin{aligned} \tilde{F}_Q &= F_C + F_Q \\ &= \sum_{\lambda} \frac{(\partial p_{\lambda}/\partial \vartheta)^2}{p_{\lambda}} + 2 \sum_{\lambda, \lambda'} \frac{(p_{\lambda} - p_{\lambda'})^2}{p_{\lambda} + p_{\lambda'}} \left| \left\langle \lambda \left| \frac{\partial \lambda'}{\partial \vartheta} \right\rangle \right|^2. \end{aligned} \quad (\text{S1})$$

The first term, F_C , cannot appear under purely unitary transformations. The second term, F_Q , the subject of this work, is the term that is relevant at a QPT at $T = 0$, where the first term, F_C , vanishes identically.

Both terms are related to different aspects of the susceptibility of ρ towards the generator $\hat{O}$. As demonstrated in Eq. (4) of the main text, F_Q is in one-to-one correspondence to the Kubo dynamic susceptibility at non-zero frequencies. In Kubo's linear response theory, a system is perturbed weakly in a time-dependent way,

generating a unitary dynamics, but—to leading order—without changing the populations of energy levels [3].

The change of probabilities, instead, is related to the static, isothermal susceptibility, as has already been pointed out within purely classical statistical physics [4]. Here, we demonstrate this relation in a quantum statistical calculation. Consider a static perturbation, $H \rightarrow H' = H + \vartheta' \hat{O}$, with $\vartheta' = T\vartheta$ (i.e., we choose the energy scale $\mathcal{J} = T$). The expectation value of the observable $\hat{O}$ is given by a derivative of the free entropy $\ln Z$, with $Z = \sum_{\lambda} \exp(-E_{\lambda}/T)$ the partition function,

$$\langle \hat{O} \rangle = T \frac{\partial \ln Z}{\partial \vartheta'} = - \sum_{\lambda} p_{\lambda} \frac{\partial E_{\lambda}}{\partial \vartheta'}. \quad (\text{S2})$$

(Here and in the following, we understand derivatives with respect to ϑ' to be evaluated at $\vartheta' = 0$.) The corresponding static, isothermal susceptibility is

$$\chi^T \equiv \frac{\partial \langle \hat{O} \rangle}{\partial \vartheta'} = - \sum_{\lambda} \frac{\partial p_{\lambda}}{\partial \vartheta'} \frac{\partial E_{\lambda}}{\partial \vartheta'} - \sum_{\lambda} p_{\lambda} \frac{\partial^2 E_{\lambda}}{(\partial \vartheta')^2}. \quad (\text{S3})$$

The derivative of p_{λ} reads

$$\frac{\partial p_{\lambda}}{\partial \vartheta'} = - \frac{p_{\lambda}}{T} \left(\frac{\partial E_{\lambda}}{\partial \vartheta'} - \sum_{\lambda'} p_{\lambda'} \frac{\partial E_{\lambda'}}{\partial \vartheta'} \right) \quad (\text{S4})$$

One can use perturbation theory to evaluate the change of energy levels following the static perturbation. Working in a basis where degenerate eigenlevels $|\lambda\rangle$ are prediagonalised with respect to the perturbation operator $\hat{O}$, we obtain

$$\begin{aligned} \chi^T &= \frac{1}{T} \left(\sum_{\lambda} p_{\lambda} \hat{O}_{\lambda, \lambda}^2 - \langle \hat{O} \rangle^2 \right) + \sum_{\lambda, \lambda'}^{E_{\lambda} \neq E_{\lambda'}} \frac{|\hat{O}_{\lambda, \lambda'}|^2 (p_{\lambda} - p_{\lambda'})}{E_{\lambda'} - E_{\lambda}} \\ &= \chi_{\text{el}} + \chi_{\text{vV}}. \end{aligned} \quad (\text{S5})$$

Here, we defined the abbreviation $\hat{O}_{\lambda, \lambda'} \equiv \langle \lambda | \hat{O} | \lambda' \rangle$. The first term is the elastic contribution, or classical Curie

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contribution, arising from a change of populations. It vanishes in a pure state, such as the quantum ground state at $T = 0$. The second term is the quantum-mechanical van-Vleck correction, arising from matrix elements between different energy levels. It is connected smoothly to the real part of the Kubo dynamic susceptibility at non-zero frequencies and vanishes in a purely classical theory. Importantly, the elastic peak is a singular contribution at exactly $\omega = 0$, and does thus not influence F_Q . Since it is this elastic peak that diverges at thermal phase transitions, these do not appear as particular features in F_Q .

We can connect the elastic peak to the classical contribution to the Fisher information [4]. Equation (S4) yields $F_C = \sum_{\lambda} \frac{(\partial p_{\lambda}/\partial \theta)^2}{p_{\lambda}} = \chi_{\text{el}}/T$. Similarly to χ^T , $\tilde{F}_Q$ thus has two parts: $\tilde{F}_Q$, the subject of this article, related to entanglement and divergent at QPTs; and F_C , of non-unitary, classical origin and divergent at thermal phase transitions. Both are related to the order parameter's susceptibility. Remarkably, by choosing the susceptibility at vanishing or non-vanishing frequencies, we can extract the purely classical or the purely quantum-mechanical contribution to the QFI.

S2. SUM RULE

Our central relation Eq. (2) allows us to derive a Thomas-Reiche-Kuhn-type [5–7] sum rule that may be useful to bound the QFI. Since $\tanh(\frac{\omega}{2T}) \leq \frac{\omega}{2T}$, we have

$$F_Q(T) \leq \frac{1}{T} \frac{2}{\pi} \int_0^{\infty} d\omega \omega \chi''(\omega, T). \quad (\text{S6})$$

Due to the Kramers–Kronig relations, we can now take advantage of a superconvergence theorem [8]. This theorem states that two complex functions g and f , for which $g(x) = P \int_0^{\infty} d\tilde{x} \frac{f(\tilde{x})}{x^2 - \tilde{x}^2}$ holds, are related via $\int_0^{\infty} dx f(x) = \lim_{x \rightarrow \infty} (x^2 g(x))$, with P the principal part of the integral. By setting $f(x) = -\frac{2}{\pi} x \chi''(x)$ and $g(x) = \chi'(x) \equiv \Re(\chi(x))$, one obtains

$$F_Q(T) \leq -\frac{1}{T} \lim_{\omega \rightarrow \infty} (\omega^2 \chi'(\omega, T)), \quad (\text{S7})$$

a Thomas-Reiche-Kuhn-type sum rule that upper bounds the QFI. Here, as in Eq. (4) of the main text, the limit to infinity is to be understood as a frequency range much larger than all energy scales of the considered degrees of freedom (but smaller than energy scales at which resonances to additional degrees of freedom appear, the entanglement between which, however, could be analysed analogously). It is interesting to note that, if we take the classical limit by sending $T \rightarrow \infty$, the bound (S7) becomes an equality relating the QFI to the high-frequency limit of $\chi'(\omega)$.

S3. SCALING AND UNIVERSALITY

In the main text, we have demonstrated a direct relationship of the QFI to thermodynamic response functions. These response functions underlie universal scaling behaviour close to quantum critical points, and consequently directly imprint a universal scaling behaviour onto the QFI. Aspects of critical scaling of the QFI and related quantum metrics have also been discussed, e.g., in Refs. [9–16]. The purpose of the present supplementary section is to pedagogically present the scaling theory for the QFI.

A. Basic principles

Before discussing the scaling of the QFI in detail, let us briefly recall the basic principles of the scaling Ansatz [17]. Here, we focus on lattice models with lattice spacing a that have a continuous phase transition at zero temperature, *viz* a quantum phase transition (QPT). Generalisation to continuum models is straightforward. The dimensionality of the system is d , its linear size L , and the total particle number $N = L^d$ (where we set $a = 1$). To simplify the discussion, we focus on the simplest scenario, QPTs obtained by tuning a single relevant field, say the magnetic field h , to the critical point h_c . Sufficiently close to the phase transition, very diverse microscopic models can develop the same collective emerging behaviour at large length scales, i.e., they display the same power-law decay of correlations, which is described by a small set of universal exponents $\{\Delta_{\alpha}\}$. Such a universality emerges between microscopic Hamiltonians if they differ only by terms (called irrelevant operators) that vanish under coarse-graining transformations on the lattice. The corresponding Hamiltonians become more and more similar in the infra-red limit, and the emerging physics becomes the same.

In finite systems, or systems at finite temperature, the power-law of the correlations is cut off by system size or the thermal correlation length. In such cases, one can extract the universal properties of the QPT by performing an appropriate analysis of scale transformations, i.e., transformations that modify the lattice spacing $a \rightarrow \lambda a$, where $\lambda > 1$ (this scale transformation can be envisioned as grouping subsets of λ lattice sites together). At the critical point one can expand all local operators in the basis of scaling operators, i.e., those that transform under rescaling with a simple power of λ , $\hat{\mathcal{O}}_i^{\alpha} \rightarrow \lambda^{-\Delta_{\alpha}} \hat{\mathcal{O}}_i^{\alpha}$, with scaling exponent $\Delta_{\alpha} \geq 0$. The two-point correlation function of these operators decays as

$$\langle \hat{\mathcal{O}}_0^{\alpha} \hat{\mathcal{O}}_r^{\alpha} \rangle \propto r^{-2\Delta_{\alpha}}. \quad (\text{S8})$$

In particular, the fixed-point Hamiltonian is a scaling operator with scaling exponent Δ_H .

Scaling operators that transform with $\Delta_{\alpha} > \Delta_H$ are called *irrelevant* operators, since, when added as a weak

perturbation, coarse graining decreases their importance relative to the original Hamiltonian. Those operators with $\Delta_\alpha < \Delta_H$, on the other hand, are called *relevant*, since they become larger and larger under coarse-graining transformations, eventually driving the system to a different fixed point. In the *scaling regime*, the regime where relevant perturbations are still small compared to the original Hamiltonian, one can extract the universal low-energy physics by studying the response of the system to scaling transformations.

Relevant perturbations are, e.g., the inverse system size L^{-1} , the finite temperature T [18], and the normalized distance from the critical point $\tilde{h} = |h - h_c|/h_c$. They increase under a rescaling as $L^{-1} \rightarrow L'^{-1} = \lambda L^{-1}$, $T \rightarrow T' = \lambda^z T$, and $\tilde{h} \rightarrow \tilde{h}' = \lambda^{1/\nu} \tilde{h}$, where z is the dynamical critical exponent and ν the correlation-length critical exponent. The scaling regime for a finite and cold system is thus defined by $L^{-1}, T \ll 1$, where we can consider inverse system size and temperature as small perturbations and project the full theory on the scaling operators, the eigenvectors of the linearized scaling transformation close to the critical point. We will now discuss the scaling behaviour of the QFI with respect to these perturbations.

B. Scaling of the quantum Fisher information

As the starting point, we first discuss the scaling of the ground-state QFI in the vicinity of a quantum phase transition [12–15]. Afterwards, we will give the generalization to non-zero temperatures. In the ground state, the QFI related to a scaling operator, $\hat{O} = \sum_l \hat{O}_l^\alpha$, takes the simple form [19]

$$f_Q = \frac{4}{L^d} \sum_l \sum_r \langle \hat{O}_l^\alpha \hat{O}_{l+r}^\alpha \rangle. \quad (\text{S9})$$

We want to understand how it transforms under scaling transformations. On the lattice, a scaling transformation of a factor $\lambda > 1$ implies grouping λ sites together in a single block. Under a scale transformation $L \rightarrow L' = L/\lambda$, $l \rightarrow l' = l/\lambda$, $r \rightarrow r' = r/\lambda$. Remembering that scaling operators transform as $\hat{O}_l^\alpha \rightarrow \lambda^{-\Delta_\alpha} \hat{O}_l^\alpha$, if we assume that the ground state is invariant under such a transformation and that the correlation function varies slowly inside each block, we obtain

$$f_Q = \lambda^{d-2\Delta_\alpha} \frac{4}{L'^d} \sum_{l'} \sum_{r'} \langle \hat{O}_{l'}^\alpha \hat{O}_{l'+r'}^\alpha \rangle. \quad (\text{S10})$$

If we now define the scaling exponent for the QFI

$$\Delta_Q = d - 2\Delta_\alpha, \quad (\text{S11})$$

the QFI transforms as

$$f_Q \rightarrow \lambda^{\Delta_Q} f_Q. \quad (\text{S12})$$

(note the similarity to the scaling of the geometric tensor of eigenstates, which has scaling exponent $\Delta_Q + 2z$ [9, 10]).

One can derive this scaling behaviour also at non-zero temperatures starting from our central formula (4), which is valid at any T . To this end, we consider the dynamic structure factor

$$S(\omega) = \frac{1}{2} \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \hat{O} \hat{O}(t) + \hat{O}(t) \hat{O} \rangle, \quad (\text{S13})$$

where $\hat{O}(t) = e^{iHt} \hat{O} e^{-iHt}$. Inserting $S(\omega)$ into Eq. (4) by use of the fluctuation–dissipation relation $\chi''(\omega) = \tanh(\omega/2T) S(\omega)$, the QFI density becomes

$$f_Q = \frac{2}{\pi L^d} \int_{-\infty}^{\infty} d\omega \tanh^2\left(\frac{\omega}{2T}\right) S(\omega). \quad (\text{S14})$$

Since under a scale transformation frequencies and temperatures scale in the same way [20], $T \rightarrow \lambda^z T$ and $\omega \rightarrow \lambda^z \omega$, the above formula has the same scaling behaviour as $2 \int_{-\infty}^{\infty} d\omega S(\omega) / (\pi L^d) = 4 \langle \hat{O}^2 \rangle / L^d$. We are thus back at Eq. (S9), and re-obtain the scaling behaviour (S12).

In a large but finite system, close to the critical point, and for temperatures close to zero, considering how $L, T, \tilde{h}$ transform under scale transformations, we thus expect

$$f_Q(T, L^{-1}, \tilde{h}) = \lambda^{\Delta_Q} \phi_Q(\lambda^z T, \lambda L^{-1}, \lambda^{1/\nu} \tilde{h}) + C. \quad (\text{S15})$$

The constant C is related to the non-universal short-distance correlations. The scaling function ϕ_Q encodes the large-distance part of the correlation functions, which dominates the scaling if the condition $\Delta_Q > 0$ is fulfilled. In the following discussion, we assume that this is the case, allowing us to neglect the non-universal constant C , leading to Eq. (7). Below we will encounter, in the guise of the hard-core boson chain, an example where $\Delta_Q \leq 0$. In such cases, derivatives of f_Q still allow one to extract the purely universal contribution. Note that when $\Delta_Q = 0$ there can be logarithmic corrections to Eq. (S15).

We can use the scaling form Eq. (S15) as long as all relevant perturbations are small, that is as long as we stay in the scaling regime. The most relevant perturbation will constitute the most significant breaking of scale invariance, a fact that enables us to extract valuable information about the involved critical exponents. For example, when both L^{-1} and T are sufficiently small, i.e., if $L^{-1}, T^{1/z} < \tilde{h}^\nu$, the deviation $\tilde{h}$ from the critical point represents the most relevant perturbation: scale invariance will be broken at the scale $\lambda \sim \tilde{h}^{-\nu}$. This implies that we can trade λ by $\tilde{h}^{-\nu}$, and varying $\tilde{h}$ in this regime has the same effect as a rescaling transformation. In this regime, we expect f_Q therefore to behave as

$$f_Q(T, L^{-1}, \tilde{h}) = \tilde{h}^{-\nu \Delta_Q} \phi_{Q1}\left(T \tilde{h}^{-\nu z}, L^{-1} \tilde{h}^{-\nu}\right). \quad (\text{S16})$$

For sufficiently relevant operators, for which $\Delta_Q > 0$, f_Q thus diverges as $\tilde{h}^{-\nu\Delta_Q}$ when approaching the critical point. In the above expression, $\phi_{Q1}(T\tilde{h}^{-\nu z}, L^{-1}\tilde{h}^{-\nu})$ is a scaling function, embodying the fact that the normalised QFI density, $f_Q/\tilde{h}^{-\nu\Delta_Q}$, is a well defined function of the scaling variables $T\tilde{h}^{-\nu z}$ and $L^{-1}\tilde{h}^{-\nu}$ rather than of L , T , and $\tilde{h}$ separately.

Upon decreasing $\tilde{h}$, the system will reach a crossover with the next relevant perturbation. If $L^{-1} < T^{1/z}$, the crossover will occur around $T^{1/z} \sim \tilde{h}^\nu$ and carry over into a thermally dominated regime, for which the scaling of the QFI is dictated by

$$f_Q(T, L^{-1}, \tilde{h}) = T^{-\Delta_Q/z} \phi_{Q2}(L^{-1}T^{-1/z}, \tilde{h}T^{-1/(z\nu)}) . \quad (\text{S17})$$

This scaling behaviour characterises the robustness of critical entanglement against non-zero temperatures (see, e.g., Ref. [21] for a discussion of the robustness in presence of noise).

In the opposite case $L^{-1} > T^{1/z}$, upon reducing $\tilde{h}$ one enters a regime dominated by finite-size effects. Here, f_Q behaves as

$$f_Q(T, L^{-1}, \tilde{h}) = L^{\Delta_Q} \phi_{Q3}(TL^z, L^{1/\nu}\tilde{h}) . \quad (\text{S18})$$

In this regime, f_Q saturates to a finite, L -dependent value, which diverges with system size as L^{Δ_Q} . The crossover is located around $L^{-1} \sim \tilde{h}^\nu$ and should thus approach the critical value $\tilde{h} = 0$ as $L^{-1/\nu}$. Another interesting scenario appears when T and $1/L$ are comparable, giving rise to further complex crossover phenomena. This can be seen, e.g., in Fig. 2 of the main text, where we study the universal scaling behaviour of the QFI in the paradigmatic model of the quantum Ising chain.

C. Multipartite entanglement length

This scaling analysis immediately permits us to identify strongly entangled quantum critical points. These are those critical points where $\Delta_Q > 0$, i.e., where at low temperatures and close to criticality, f_Q diverges with system size. Additionally, the scaling also allows us to extract a length scale of the entanglement present in the system [22]. Combining Eq. (S15) and main-text inequality (1), one sees that the number of entangled particles scales as $N_Q \geq f_Q \sim \lambda^{d-2\Delta_Q}$, which defines a multipartite entanglement length scale

$$l_{\text{ent}} \sim N_Q^{1/d} \gtrsim \lambda^{1-2\Delta_Q/d} . \quad (\text{S19})$$

Here, we assumed an isotropic distribution of entanglement; for anisotropic situations, the longest distance along which particles are witnessed to be entangled is actually larger. This entanglement length is related to the correlation length ξ of the generator $\hat{O}$. Since at criticality the correlation length scales as $\xi \sim \lambda$, one has $l_{\text{ent}} \sim \xi^{1-2\Delta_Q/d}$. For $2\Delta_Q/d < 1$, the length scale over

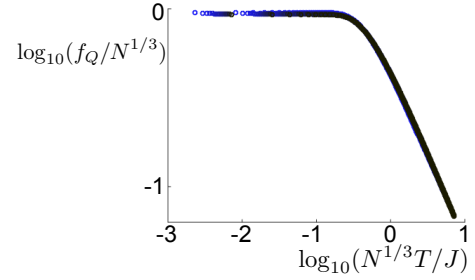


Figure S1. Universal scaling of the QFI density, $f_Q = F_Q/N$, calculated for the order parameter in the quantum Ising model with infinite-range interactions. Finite-temperature and finite-size scaling at criticality $\gamma = \pi/4$. The bullets denote exact data, from lighter to darker blue for $N = 100, \dots, 3000$. The universal scaling laws produce an excellent data collapse. The divergence $f_Q \sim N^{1/3}$ is much weaker than in the one-dimensional quantum Ising model, reflecting the more mean-field-like character of the infinite-range model. Nevertheless, it still implies a diverging multipartiteness of entanglement.

which particles are entangled with each other thus diverges with system size.

S4. EXAMPLES FOR UNIVERSAL SCALING

In the main text, we illustrate the universal scaling of the QFI using one of the most simple and paradigmatic models hosting a quantum phase transition, the Ising model in a transverse field. In this section, we spotlight additional aspects of the universal behaviour of the QFI, exemplified in three different models. First, we briefly provide the scaling analysis for the quantum Ising model with infinite-range interactions capturing the universal properties of the Ising transition above the upper critical dimension (Sec. S4 A). Then, we study the Mott-insulator–superfluid transition of hard-core bosons to show that relevant operators other than the order parameter also exhibit universal behaviour (Sec. S4 B). Finally, we analyze the QFI in the Kitaev wire, a model that has a topological quantum phase transition (Sec. S4 C). We show that while the QFI does not witness strong entanglement in this model, it is nevertheless sensitive to the topological transition.

A. The infinite-range quantum Ising model

In the main text, we considered the infinite-range Ising model in a transverse field, main-text Eq. (8) with $\mathcal{I}_j = 1/N$, to illustrate the absence of features in the QFI at thermal transitions (see Fig. 3). Here, we provide its scaling analysis in the vicinity of its quantum phase transition at non-zero temperatures, using exact numerical results (see Methods). Scaling behaviour of this model in its ground-state has been discussed, e.g., in

Refs. [12, 23].

The infinite-range Ising model describes the universal properties above the upper critical dimension, with the longitudinal magnetization $\hat{O} = \sum_{l=1}^N \sigma_l^x / 2$ as the order parameter. In Fig. S1, we plot the QFI as a function of temperature for different system sizes at the system's quantum critical point, $\tilde{h} = 0$. Using that $\Delta_Q = 1/3$ and $z = 1/3$ [24] one obtains the scaling of the QFI density $f_Q \sim N^{1/3} \phi(TN^{1/3})$. From the scaling collapse, we determine numerically that $\Delta_Q = 0.34$ and $z = 0.33$, which is in very good agreement with the exact solution.

B. Mott-insulator–superfluid transition of hard-core bosons

The scaling behaviour of f_Q , Eq. (S15), appears in all relevant operators, not only the order parameter. As an illustration for this fact, we choose a system that is relevant for experiments on cold gases, namely a system of ultracold bosonic atoms confined in a one-dimensional optical lattice. In the limit of strong on-site interactions, this system can be modelled by the Hamiltonian [25]

$$H_{\text{hcb}} = -J \sum_{l=1}^{N-1} (\hat{b}_l^\dagger \hat{b}_{l+1} + \text{h.c.}) - 2\mu \sum_{l=1}^N \hat{b}_l^\dagger \hat{b}_l, \quad (\text{S20})$$

where the bosonic creation (annihilation) operators $\hat{b}_l^\dagger$ ($\hat{b}_l$) obey the hard-core constraint $\hat{b}_l^\dagger \hat{b}_l^\dagger = 0$. This model has a QPT at $\mu_c/J = 1$ from a critical superfluid phase to a Mott-insulator at uniform filling, with critical exponents $z = 2$ and $\nu = 1/2$ [20].

The transition has several peculiarities. First, the exact ground state is fully separable at the QPT [26]. Second, one of its order parameters, the particle density $\hat{n} = \sum_l \hat{b}_l^\dagger \hat{b}_l / N$, commutes with the Hamiltonian. To get a non-zero signal in the QFI, we can, however, use a modified observable that is still a relevant operator, the staggered density $\hat{O} = \sum_l (-1)^l \hat{b}_l^\dagger \hat{b}_l$. In optical-lattice experiments, the dynamic susceptibility of such density fluctuations may be probed by Bragg spectroscopy [27, 28].

The Hamiltonian Eq. (S20) is exactly solvable with the same methods as for the Ising chain in a transverse field [20, 29]. Moreover, under periodic boundary conditions, the dynamic structure factor [see Eq. (S13)] associated to density fluctuations $\hat{O} = \sum_l e^{iqr_l} \hat{b}_l^\dagger \hat{b}_l$ at wave vector q can be evaluated analytically [30]. Via the fluctuation–dissipation relation, the dynamic structure factor in turn determines the dynamic susceptibility. Specifying the argument q explicitly, one finds that $S(q, \omega)$ is gapless at $q = 0$ and acquires a maximum at $q = \pi$, with

$$S(\pi, \omega) = 2\pi \sum_k \delta(\omega - \epsilon_k + \epsilon_{k+\pi}) f(\epsilon_{k+\pi}) [1 - f(\epsilon_k)], \quad (\text{S21})$$

with $\epsilon_k = -2[J + \mu \cos(k)]$ the single-particle energy of the diagonalised Hamiltonian and $f(\epsilon_k) =$

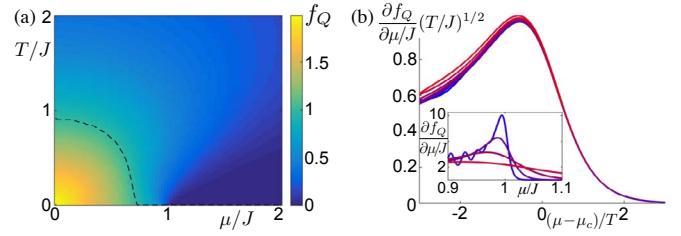


Figure S2. Quantum Fisher information density, f_Q , for the staggered density of hard-core bosons in a one-dimensional optical lattice. (a) In part of the superfluid phase, f_Q witnesses two-site entanglement. Exact data for $N = 1000$. (b) Inset: $\partial f_Q / \partial \mu$ shows a peak that, with decreasing T , diverges and moves towards the critical point. Main panel: The predicted scaling exponents produce a perfect data collapse for $\partial f_Q / \partial \mu$. Exact data for $N = 100$ and, from blue to red, 8 different temperatures distributed on a logarithmic scale between $T = 0.01 - 0.2J$ (inset: 4 temperatures in $0.01 - 0.13J$).

$1/[\exp(\epsilon_k/T) + 1]$ the Fermi–Dirac occupation. As a consequence of the gaplessness, the QFI evaluated at $q = 0$ vanishes, while it assumes its maximum value at $q = \pi$, i.e., for the staggered density. Using the fluctuation–dissipation theorem and evaluating the integral in Eq. (4), the corresponding QFI density becomes

$$f_Q = \frac{4}{N} \sum_k \tanh^2 \left(\frac{\epsilon_k - \epsilon_{k+\pi}}{2T} \right) f(\epsilon_{k+\pi}) [1 - f(\epsilon_k)]. \quad (\text{S22})$$

This quantity is plotted in Fig. S2(a) as a function of temperature and chemical potential, for $N = 1000$. The entanglement witnessed by the QFI extends up to $T \approx 0.91J$, improving over other witnesses showing entanglement up to $T \approx 0.7J$ [31, 32]. In contrast to Ref. [31], however, the QFI for the staggered magnetisation does not witness genuine multipartiteness of entanglement, but saturates at the value $f_Q = 2$. (Notably, the QFI for other generators such as $b_l + b_l^\dagger$ does show divergent behaviour.) Towards the phase transition, the QFI for the staggered density decreases, consistent with the presence of the factorisation point. Despite this absence of divergences, the phase transition imprints its character onto the universal behaviour of the QFI and its derivatives, as explained in what follows.

At $T = 0$ and $N = \infty$, one can use Eq. (2) to derive a particularly simple analytical formula for the QFI of the staggered density,

$$f_Q = \begin{cases} 4n & \mu \leq 0 \\ 4(1-n) & \mu > 0 \end{cases}. \quad (\text{S23})$$

Here, we defined the mean particle density $n \equiv \langle \hat{n} \rangle$, which is $n = 1 - \text{acos}(\mu/J)/\pi$ for $|\mu| < J$ and 0 else. The QFI density attains its maximal value of $f_Q = 2$ at $\mu = 0$ and decreases towards the phase-transition point, where it vanishes. The above analytic expression allows us to obtain the scaling dimension of f_Q by expanding it close to the critical point, $f_Q \propto \tilde{h}^{1/2}$,

with $\tilde{h} = 1 - \mu/J$ and $\mu/J \leq 1$. Comparison with the scaling form Eq. (S16) gives the exact scaling exponent $\Delta_Q = -1$. Since $\Delta_Q < 0$, f_Q of the staggered magnetisation displays no divergent behaviour.

Nevertheless, as seen in the inset of Fig. S2(b), the non-analyticity of f_Q at $\tilde{h} = 0$ leads to a divergence in its derivatives. Considering the temperature as the most relevant perturbation, Eq. (S17), we obtain the universal behaviour

$$\frac{\partial f_Q}{\partial \tilde{h}} = (T/J)^{-\Delta_Q/z+1/(z\nu)} \phi_{Q3} \left(\tilde{h} (T/J)^{-1/(z\nu)} \right). \quad (\text{S24})$$

That is, $\partial f_Q / \partial \tilde{h}$ has a peak that diverges as $(T/J)^{-\Delta_Q/z+1/(z\nu)}$ and the position of which lies at a line $\tilde{h} \sim (T/J)^{1/(z\nu)}$. Using the analytically known exponents $-\Delta_Q/z + 1/(z\nu) = 1/2$ and $1/(z\nu) = 1$, an excellent scaling collapse is achieved up to rather large temperatures [main panel of Fig. S2(b)]. We also performed fits to the numerical data for the peak height and peak position. These yield exponents of 0.49 and 1.02, respectively—even without high-precision data, we are able to reproduce the exact scaling exponents within two percent accuracy.

C. Topological transition in the Kitaev wire

Entanglement, in particular the entanglement entropy, has proven important for the characterization of topological systems [33]. The natural question thus arises whether the QFI as an entanglement witness is capable to quantify and characterize topological phases or topological phase transitions. On a fundamental level, topological order cannot be captured by local observables that appear in conventional dynamic susceptibilities such as considered for the QFI in the main text. Still, the QFI is, remarkably, capable to detect topological phase transitions, as we will see in the example of the Kitaev chain studied in the following. It will be an interesting research question whether the QFI or non-local variations thereof might be useful for quantifying topological phases.

One of the simplest examples of a topological transition is given by the Kitaev wire [34], a one-dimensional model of free fermions with both tunnelling and pairing,

$$H_{\text{Kw}} = -J \sum_{l=1}^{N-1} \left(\hat{c}_l^\dagger \hat{c}_{l+1} + \Delta \hat{c}_l^\dagger \hat{c}_{l+1}^\dagger + \text{h.c.} \right) + 2\mu \sum_{l=1}^N \hat{c}_l^\dagger \hat{c}_l. \quad (\text{S25})$$

Here, a fermion at site l is created (annihilated) by $\hat{c}_l$ ($\hat{c}_l^\dagger$). Being a free-fermion model, this model can be solved exactly [35, 36]. For $\Delta = 1$, its eigenenergies coincide with those of the Ising chain in a transverse field [20], which is studied in the main text. Importantly, however, although the spectra are identical, the Jordan–Wigner transformation that establishes the mapping is highly non-local. Thus, operators or correlation functions that are local

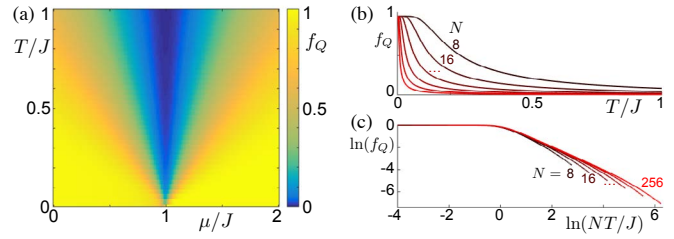


Figure S3. Quantum Fisher information density, f_Q , for non-interacting fermions in a one-dimensional Kitaev wire with equal tunnelling and pairing, for the generator $\hat{O} = \sum_i (\hat{c}_i + \hat{c}_i^\dagger)/2$. (a) f_Q does not witness entanglement, but it is sensitive to the presence of the topological quantum critical point. Exact data for $N = 128$. (b) Cuts along temperature, at the critical point, $\mu/J = 1$. The sensitivity of f_Q increases with system size. (c) A perfect data collapse is achieved with the predicted critical exponents. Deviations appear where T leaves the critical regime, i.e., where the non-universal high-temperature behaviour sets in. Data in panels b and c, from dark to light red: $N = 8, 16, \dots, 256$.

in the Ising chain do not necessarily transform into local objects in the Kitaev model and vice versa. For example, the conventional symmetry-broken phase in the Ising model can on the Kitaev side only be captured by string order parameters, i.e., by non-local correlation functions. This phase is separated from a trivial phase at large chemical potential by a topological transition.

In the following, we set the pairing strength $\Delta = 1$ and tune the chemical potential μ to access the topological transition, choosing the generator $\hat{O} = \sum_i (\hat{c}_i + \hat{c}_i^\dagger)/2$ to study it. This operator is only local for the Kitaev chain; if we were to use the Jordan–Wigner transformation to convert into spin language, it would become a sum of non-local operators. Thus, the associated dynamic susceptibility defining the QFI is only meaningful in the Kitaev model but not the Ising chain. Figure S3 presents the corresponding QFI density in the chemical-potential–temperature plane. Its value does not exceed $f_Q = 1$; contrary to other cases analyzed above, the QFI does not witness entanglement. Importantly, however, as Fig. S3 shows, the QFI is able to detect the presence of the topological phase transition in the Kitaev wire. This behaviour is not only captured by the considered observable, but also by other generators not shown here.

What is more, the QFI displays clear universal behaviour, which we analyze analogous to Fig. 2 of the main text, i.e., by studying cuts along T for different system sizes, with the chemical potential fixed to the critical value. Since the transition is of Ising type, it has the critical exponents $\nu = z = 1$. Additionally, as shown by Fig. S3(b), the QFI density approaches $f_Q = 1$ at $T = 0$, i.e., we have $\Delta_Q = 0$. In consequence, we expect from Eq. (S17) or (S18) f_Q to be a universal function of TN within in the critical regime. This expectation is confirmed by the scaling collapse displayed in Fig. S3(c).

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